NETWORKING

Forward & Reverse Proxies

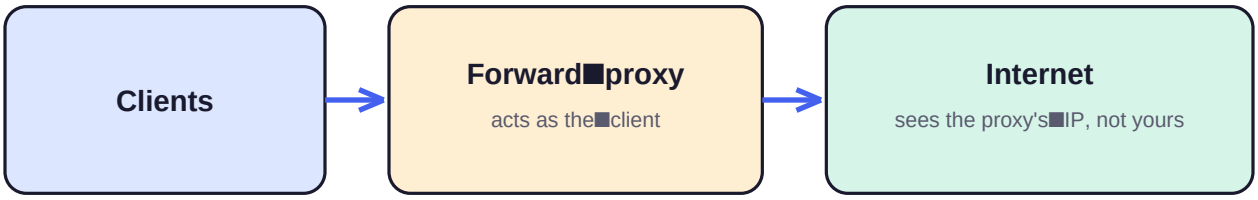
The middlemen of the web —
who they serve, and what they rewrite.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Forward Proxy: For the Client

Sits in front of CLIENTS, fetching on their behalf.

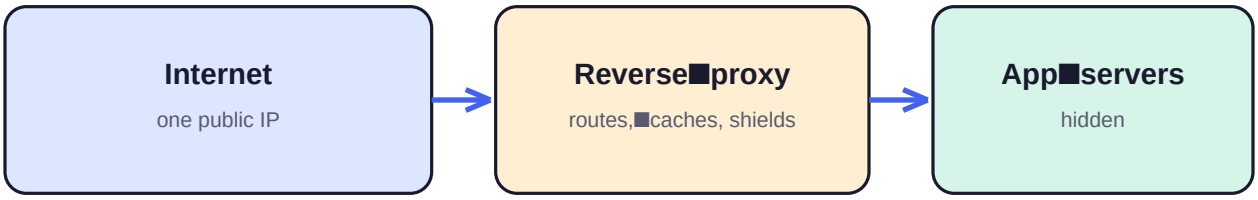


- Use cases: corporate web filtering, caching, anonymity, bypassing geo-blocks.
- The server never sees the real client — the proxy's IP is the source.
- CONNECT method tunnels TLS through it; the proxy sees the destination, not the content.

STEP 2/5

Reverse Proxy: For the Server

Sits in front of SERVERS, presenting one face to the world.



- Use cases: TLS termination, caching, request routing, hiding backend topology.
- Clients never learn backend IPs — the proxy is the only public surface.
- nginx, HAProxy, Envoy, Cloudflare: all primarily reverse proxies.

STEP 3/5

Header Surgery: X-Forwarded-For

Proxies rewrite identity — headers carry the truth (if you trust them).

```
X-Forwarded-For: 203.0.113.7, 198.51.100.23

X-Forwarded-Proto: https

Forwarded: for=203.0.113.7; proto=https  (RFC 7239, the standard)
```

- Each proxy APPENDS the client IP it saw — leftmost is the original client (usually).
- Spoofable: anyone can inject X-Forwarded-For — only trust it from YOUR proxies (strip at the edge).
- Apps behind proxies must read these headers for real IPs, correct redirects (http→https), and rate limiting.

STEP 4/5

Transparent Proxy

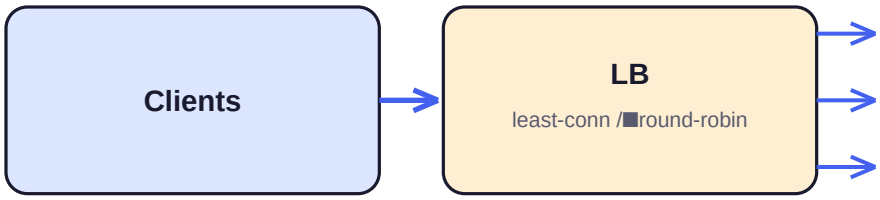
Intercepts traffic without the client knowing — or consenting to config.

- Deployed by ISPs and enterprises: port-80/443 traffic steered via routing or WCCP.
- HTTPS interception requires a trusted CA on the client — effectively a sanctioned MITM.
- Controversial for privacy; common for malware scanning and data-loss prevention.

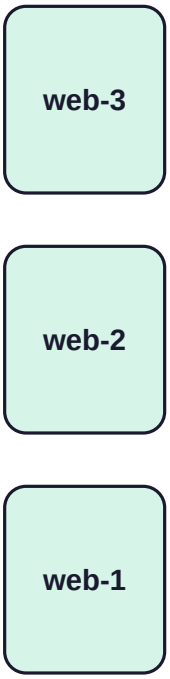
STEP 5/5

Load Balancer as Reverse Proxy

Every load balancer is a reverse proxy with a scheduling algorithm.



- Health checks: dead backends are pulled from rotation automatically.
- Sticky sessions: route the same client to the same backend when state is local.
- L4 (TCP) vs L7 (HTTP): L7 can route by path/host header; L4 just spreads connections.



STEP ✓/✓

Key Takeaways

Proxies — Remember

- Forward proxy serves clients (filtering, caching, anonymity); reverse proxy serves servers (TLS, routing, shielding).
- X-Forwarded-For carries the real client IP — append-only, spoofable, trust only your own proxies.
- Transparent proxies intercept without client config — powerful, privacy-sensitive.
- A load balancer is a reverse proxy that distributes: health checks, algorithms, L4 vs L7.
- The proxy is a trust boundary: it sees destinations (forward) or full requests (reverse) — choose operators accordingly.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev